



Figure 1: The template



Figure 2: Template with placeholder text added

layers, one labelled 'Background' and the other with `<name>`. If it is so, you are on the right track.

Now save the file with `Ctrl + S`, and exit GIMP.

We now need to get the file with the participants' names ready. Any format can be used, but in this case, we will use a normal text file. Enter all names, one per line and save the file.

Here comes the coding part. The code given below comes with explanations given as comments (beginning with #):

```
#!/usr/bin/env python
from gimpfu import *
# this is the important module for coding in python GIMP
import os # for file/folder paths
def run(*args):
    template_file, list_file, output_folder = args
    # the data is sent is unpacked into 3 variables
    # this is where we will store our auto-generated certificates
    output_folder = os.path.expanduser('~/' + output_folder)
    if not os.path.exists(output_folder):
        os.makedirs(output_folder)
    # create the output directory, if its not there already
    # this number is used for avoiding collisions when
    # 2 people have the same name
    # if there are 2 people named X,
    # one file will have a name containing X1, the other X2 etc.
    i = 0
    with open(list_file, 'r') as list: # open the file for reading
        for name in list:
```

```
    # for each line do something i.e for each name in the list
        i += 1
        bg_image = pdb.gimp_file_load(template_file, template_file)
        # load the certificate
        # create a reference to the name layer in the certificate
        name_layer = filter(lambda x: x.name == '<name>', bg_
            image.layers)[0]
        # change below 2 values in brackets if needed
        pdb.gimp_text_layer_set_font(name_layer, 'Comic Sans MS')
        pdb.gimp_text_layer_set_font_size(name_layer, 20, 0)
        # set the required name in the certificate
        pdb.gimp_text_layer_set_text(name_layer, name)
        # We can save only one layer at a time, so merge them
        merged = pdb.gimp_image_merge_visible_layers(bg_image, 0)

        # save file as <name><some number>
        output_filename = name + str(i) + ".png"
#relative path
        output_filename = os.path.join(output_folder,
            output_filename)
#absolute path

        # save the file as a PNG image
        pdb.file_png_save_defaults(bg_image, merged,
            output_filename, output_filename)
        print "Finished"
```

```
# this is an important part, without this,
# the plug-in will not appear in GIMP
# first line is the name that appears in the title bar
# second line is the menu hierarchy
# the '[' starting from the third line provides the selection
# tools for selecting the files/directories we # need
# the last line calls the run() method defined above
register(
    "certificate_generator", "", "", "", "",
    "<Toolbox>>Xtns/Languages/Python-Fu/_Certificate-Generator",
    "",
    [
        (PF_FILE, "arg0", "Certificate Template file", ""),
        (PF_FILE, "arg1", "Data File", ""),
        (PF_STRING, "arg2", "Output directory (created relative
            to user home)", ""),
    ],
    [],
    run
)

main()
```

Save the script as `<your_script_name>.py`. There are a number of places to save it but the one that worked out for me was `/usr/lib/gimp/2.X/plugin-ins`. Here, 2.X is the GIMP version installed in your system.